

ARender 2026.0.0

Reference benchmark — ~32,000 documents/hour sustained on a three-node Kubernetes infrastructure | Gatling, one-hour LOAD run

TEST DATE	VERSION	TOOL	DURATION	TEST SCOPE
April 28, 2026	2026.0.0	Gatling	1 hour per scenario	Rendition backend only

PREPARED BY

Petra — Uxopian Software

VERDICT: STABLE — RECOMMENDED FOR PRODUCTION



On the reference infrastructure (three-node Linux Kubernetes cluster, 8 vCPU / 32 GB RAM per worker), the ARender 2026.0.0 Rendition backend sustains **~32,000 document opens per hour at 0% error rate**, with global P99 at 240 ms. A scaling-validation run at doubled concurrency confirms near-linear headroom for higher document rates on the same infrastructure profile. The benchmark covers the Rendition backend only — UI tier, document connector, and DMS need to be sized separately in a real deployment.

1 Infrastructure & Test Scenario

REFERENCE INFRASTRUCTURE

Type	Production Linux Kubernetes cluster
Nodes	1 control plane + 3 workers
Server profile	8 vCPU, 32 GB RAM per worker node

TEST CONFIGURATION

Tool	Gatling
Test type	LOAD
Duration	1 hour per run
Scope	ARender Rendition backend (broker + converter + renderer + text handler)

TEST SCENARIO — PER-DOCUMENT REQUEST SEQUENCE

- Step 1 Open_document_with_url — Open a document via URL
- Step 2 Get_document_layout — Retrieve document layout (page count, dimensions)
- Step 3 Get_3_images — Fetch 3 rendered page images
- Step 4 Get_20_thumbnails — Fetch 20 thumbnail images
- Step 5 Get_3_pages_content — Retrieve text/content for 3 pages
- Step 6 Evict_document — Evict the document from cache

Injection mode: CLOSE. Two consecutive one-hour runs were executed: a reference run sustaining ~32,000 documents/hour, and a scaling-validation run at doubled concurrency on the same infrastructure to verify near-linear scaling.

2 Key Performance Indicators

Reference benchmark — Rendition backend on the reference infrastructure

~32,000

DOCUMENTS SERVED / HOUR

0%

ERROR RATE

8.87 /s

DOCUMENTS / SECOND

278 /s

HTTP THROUGHPUT

1,001,652

TOTAL HTTP REQUESTS

19 ms

GLOBAL P50

150 ms

GLOBAL P95

240 ms

GLOBAL P99

Scaling validation — same infrastructure, doubled concurrency

A second one-hour run was executed on the same infrastructure with doubled concurrency to verify near-linear scaling of the Rendition backend. This is a proportionality check, not a separate reference point.

~57,000

DOCUMENTS/HOUR (+77%,
NEAR-LINEAR)

0%

ERROR RATE (EFFECTIVE)

252 ms

GLOBAL P99 (VS 240 MS
REFERENCE)

155 ms

GLOBAL P95 (VS 150 MS
REFERENCE)

3 Gatling Results — Standalone / 3 nodes

Reference benchmark

Request	Total	OK	KO	% KO	Cnt/s	Min (ms)	P50 (ms)
All Requests	1,001,652	1,001,652	0	0%	278	6	19
Open_document_with_url	31,966	31,966	0	0%	8.87	14	18
Get_document_layout	31,966	31,966	0	0%	8.87	80	185
Get_3_images	53,278	53,278	0	0%	14.79	24	90
Get_20_thumbnails	223,774	223,774	0	0%	62.11	9	30
Get_3_pages_content	628,702	628,702	0	0%	174.49	6	16
Evict_document	31,966	31,966	0	0%	8.87	6	11

Scaling validation — doubled concurrency

Request	Total	OK	KO	% KO	Cnt/s	Min (ms)	P50
All Requests	1,775,943	1,775,942	1	0.00006%	493.04	6	
Open_document_with_url	56,680	56,679	1	0.002%	15.74	13	
Get_document_layout	56,679	56,679	0	0%	15.74	82	
Get_3_images	94,465	94,465	0	0%	26.23	23	
Get_20_thumbnails	396,753	396,753	0	0%	110.15	8	
Get_3_pages_content	1,114,687	1,114,687	0	0%	309.46	6	
Evict_document	56,679	56,679	0	0%	15.74	6	

Error rate — 0% effective on the reference benchmark

The reference benchmark serves 1,001,652 HTTP requests with zero errors. The scaling-validation run sustains a 0% effective error rate at doubled concurrency. All pods remained stable throughout both runs.

Document throughput — ~32,000/hour (reference) with confirmed near-linear scaling

The reference benchmark sustains 8.87 docs/s (~31,932 docs/hour) over a continuous one-hour run. The scaling-validation run confirms the architecture scales near-linearly to ~15.74 docs/s (~56,664 docs/hour) on the same infrastructure when concurrency is doubled.

Global Latency — P95 < 160 ms, P99 < 255 ms across both runs

P50, P95, and P99 remain virtually identical between the reference and scaling-validation runs (P95: 150 ms vs 155 ms; P99: 240 ms vs 252 ms), confirming excellent latency stability under doubled load.

Get_document_layout — P99 below 400 ms

P99 stays comfortably below 400 ms (377 ms reference, 346 ms scaling-validation). Maximum outlier values reflect cold-start or cache-miss events at test ramp-up; the bulk of requests complete within sub-200 ms typical latency.

Open_document_with_url — Fast and stable

Consistently low latency across both runs: P50 18–19 ms, P95 29–35 ms, P99 40–66 ms. Max below 514 ms. This critical path endpoint performs excellently.

Get_3_pages_content — Highest volume endpoint

Over 628k requests (10u) and 1.1M requests (20u) with 0 errors. P95 46–61 ms, P99 98–108 ms. Very fast content delivery confirmed at scale.

Per-endpoint latency breakdown — Reference benchmark

Endpoint	P50 (ms)	P75 (ms)	P95 (ms)	P99 (ms)	Max (ms)	StdDev	As
Open_document_with_url	18	21	29	40	400	7	c
Get_document_layout	185	214	279	377	14,631	180	c
Get_3_images	90	206	260	308	715	82	c
Get_20_thumbnails	30	42	69	124	490	21	c

Endpoint	P50 (ms)	P75 (ms)	P95 (ms)	P99 (ms)	Max (ms)	StdDev	Assessment
Get_3_pages_content	16	21	46	98	5,437	17	Good
Evict_document	11	13	25	53	473	10	Good

Per-endpoint latency breakdown — Scaling validation

Endpoint	P50 (ms)	P75 (ms)	P95 (ms)	P99 (ms)	Max (ms)	StdDev	Assessment
Open_document_with_url	19	22	35	66	514	11	Good
Get_document_layout	182	209	266	346	7,579	161	Good
Get_3_images	113	222	306	419	807	94	Good
Get_20_thumbnails	34	49	97	148	5,172	29	Good
Get_3_pages_content	18	25	61	108	551	19	Good
Evict_document	12	16	30	79	391	12	Good

5 Resource Utilization

Reference benchmark

CPU

~40%

Node-level usage, steady-state.
~2 cores consumed across all pods (renderer dominant).

OK

MEMORY

~10 GB used

Node memory stable. Viewer ~500–700 MB RSS, Broker flat, Renderer 0.5–1.5 GB RSS.

OK

NETWORK

~400 Mbit/s out

~200 Mbit/s in. Stable throughput throughout the 1-hour run.

OK

Scaling validation — doubled concurrency

CPU

~60%

Node-level usage, steady-state.
~3–4 cores consumed.

OK

MEMORY

~12–15 GB used

Renderer RSS 1–2 GB. Node
memory pressure remains within
healthy bounds at this load level.

OK

NETWORK

~700 Mbit/s out

~300–400 Mbit/s in. Bandwidth
drops briefly around 16:40 (test
artifact) then recovers.

OK

CPU — Within capacity in both runs

CPU usage scales proportionally from ~40% (reference) to ~60% (scaling validation). The worker nodes have sufficient headroom at both throughput levels.

Memory — Stable across both runs

Memory usage tracks the load level cleanly. The renderer is the dominant consumer; pod-level RSS stays within configured limits throughout the test.

Network — No bottleneck observed

Network bandwidth is well within capacity at both load levels. Throughput is stable for the duration of each run.

6 Recommendations

1

Monitor `Get_document_layout` max outlier across releases

The maximum response times on this endpoint (14,631 ms reference, 7,579 ms scaling validation) reflect cold-start or cache-miss events at ramp-up. Tracking this maximum across releases provides a useful regression indicator. An alerting threshold around 5,000 ms is a reasonable starting point.

2

Run a higher-throughput stress test on the same infrastructure

The reference and scaling-validation runs show near-linear scaling with stable latency. A run targeting ~100,000 documents/hour is the natural next step to identify where this infrastructure profile saturates (CPU, memory, or latency degradation).

EXECUTIVE SUMMARY

On the reference infrastructure (three-node Linux Kubernetes cluster, 8 vCPU / 32 GB RAM per worker), the ARender **2026.0.0** Rendition backend sustains **~32,000 document opens per hour** (8.87 docs/s, 278 HTTP req/s, P95 150 ms, P99 240 ms, 0% error rate over 1,001,652 requests) for the full one-hour run. A scaling-validation run at doubled concurrency on the same infrastructure confirms near-linear scaling, with global P95 and P99 effectively flat (155 ms / 252 ms). Resource utilisation (CPU, memory, network) stays well within the capacity of the worker nodes. The version is **recommended for production** at the tested load level. This benchmark covers the Rendition backend only — UI tier, document connector and DMS need to be sized separately in a real deployment.